

Operators and Expressions

**SIMATS Online Programming Lab**

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192312139RD

QUESTIONS 8 / 8 completed

Swap 2 No's

SWAP 2 NO'S

C Program to Swap Two Numbers.
Swapping two numbers in C programming means swapping the values of two variables.
For example, there are two variables m & n. Value of m is "2" & value of n is "3".
Before Swapping: m value = 2; n value = 3
After Swapping: m value = 3; n value = 2

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 int main()
3 {
4     int a,b,temp;
5     printf("Enter two Numbers: ");
6     scanf("%d %d", &a,&b);
7     temp = a;
8     a = b;
9     b = temp;
10    printf("After swapping:\n");
11    printf("First Number = %d\n", a);
12    printf("Second Number = %d\n", b);
13    return 0;
14 }
```

OUTPUT

Enter two Numbers: After swapping:
First Number = 20
Second Number = 10

INPUT

10
20

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QUESTIONS 8 / 8 completed

Arithmetic

ARITHMATIC

Write a program that takes two numbers as input and prints their sum, difference, product, and quotient.

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 int main ()
3 {
4     int a = 2;
5     int b = 4;
6     int sum = a + b;
7     printf ("%d ",sum);
8     int diff = b - a;
9     printf ("%d ",diff);
10    int multi = a * b;
11    printf ("%d ", multi);
12    int quo = a / b;
13    printf ("%d ", quo);
14    return 0;
15 }
```

OUTPUT

6 2 8 0

INPUT

Your INPUT goes here!



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QUESTIONS 8 / 8 completed

Convert

CONVERT

Write a program that converts Celsius to Fahrenheit. The formula is Fahrenheit = (Celsius * 1.8) + 32.

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 int main()
3 {
4     float celsius, Fahrenheit;
5     printf("Enter temperature in celsius: ");
6     scanf("%f", &celsius);
7     Fahrenheit = (celsius * 1.8) + 32;
8     printf("celsius to Fahrenheit is: %.2f\n", Fahrenheit);
9     return 0;
10 }
```

OUTPUT

Enter temperature in celsius: celsius to Fahrenheit is: 77.00

INPUT

25



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QUESTIONS 8 / 8 completed

Rectangle

RECTANGLE

Write a program that calculates the area of a rectangle given its length and width.
Problem Description: Write a program that calculates the area of a rectangle given its length and width. The program should prompt the user to enter the length and width of the rectangle as input, then calculate the area

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 int main()
3 {
4     float celsius, Fahrenheit;
5     printf("Enter temperature in celsius: ");
6     scanf("%f", &celsius);
7     Fahrenheit = (celsius * 1.8) + 32;
8     printf("celsius to Fahrenheit is: %.2f\n", Fahrenheit);
9     return 0;
10 }
```

OUTPUT

Enter temperature in celsius: celsius to Fahrenheit is: 77.00

INPUT

25

QUESTIONS 8 / 8 completed

ODD EVEN

ODD EVEN

Write a program that checks whether a number is even or odd.

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 int main()
3 {
4     int num;
5     printf("Enter the value: \n");
6     scanf("%d", &num);
7     if (num % 2 == 0)
8         printf("%d is Even\n", num);
9     else
10        printf("%d is Odd\n", num);
11    return 0;
12 }
```

OUTPUT

```
Enter the value:
10 is Even
```

INPUT

10

QUESTIONS 8 / 8 completed

Leap

LEAP

Write a program that determines whether a year is a leap year or not. A year is a leap year if it is divisible by 4, but not divisible by 100 unless it is also divisible by 400.

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 int main()
3 {
4     int year;
5     printf("Enter a Year:\n");
6     scanf("%d", &year);
7     if ((year % 400 == 0) || (year % 4 == 0 && year % 100 != 0))
8         printf("%d is a Leap Year:\n", year);
9     else
10        printf("%d is not a Leap Year:\n", year);
11    return 0;
12 }
```

OUTPUT

```
Enter a Year:
2028 is a Leap Year:
```

INPUT

2028

QUESTIONS 8 / 8 completed

Max-Min

MAX-MIN

Write a program that takes three integers as input and prints the maximum and minimum of the three.

PROGRAM EDITOR

C

Run

Save

INPUT

10
25
30

```
1 #include <stdio.h>
2 int main ()
3 {
4     int a,b,c,max,min;
5     printf("Enter three integer value: \n");
6     scanf("%d %d %d", &a, &b, &c);
7
8     max = a;
9     if (b > max)
10         max = b;
11     if (c > max)
12         max = c;
13
14     min = a;
15     if (b < min)
16         min = b;
17     if (c < min)
18         min = c;
19     printf("Maximum value: %d\n", max);
20     printf("Minimum value: %d\n", min);
21     return 0;
```

OUTPUT

```
Enter three integer value:
Maximum value: 30
Minimum value: 10
```

QUESTIONS 8 / 8 completed

Sum-Avg

SUM-AVG

Write a program that takes a list of numbers as input and computes the sum and average of the numbers.

PROGRAM EDITOR

C

Run

Save

INPUT

5
10 20 30 40 50

```
1 #include <stdio.h>
2 int main()
3 {
4     int n,i;
5     float sum = 0,avg;
6     printf("Enter the number of elements: \n");
7     scanf("%d", &n);
8     float num[n];
9     printf("Enter numbers %d: \n", n);
10    for (i = 0; i < n; i++){
11        scanf("%f", &num[i]);
12        sum += num[i];
13    }
14    avg = sum / n;
15    printf("sum = %.2f\n", sum);
16    printf("Average = %.2f\n", avg);
17    return 0;
18 }
```

OUTPUT

```
Enter the number of elements:
Enter numbers 5:
sum = 150.00
Average = 30.00
```

Decision Macking



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QUESTIONS 8 / 8 completed

+ OR -

+ OR -

C program to Check Whether a Number is Positive or Negative.

Problem Description
The program takes the given integer and checks whether the integer is positive or negative.

Testcases:
-5
e
65
..

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 int main()
3 {
4     int n;
5     scanf("%d", &n);
6     if (n > 0)
7         printf("Positive");
8     else if (n < 0)
9         printf("Negative");
10    else
11        printf("Zero");
12    return 0;
13 }
```

OUTPUT

Positive

INPUT

65



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QUESTIONS 8 / 8 completed

Greatest

GREATEST

Write a program that takes two numbers as input and determines which one is larger.

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 int main()
3 {
4     int a,b;
5     printf("Enter two numbers: \n");
6     scanf("%d %d", &a, &b);
7     if (a > b)
8         printf("%d is the greatest.\n", a);
9     else if (b > a)
10        printf("%d is the greatest.\n", b);
11    else
12        printf("Both numbers are equal.\n");
13    return 0;
14 }
```

OUTPUT

Enter two numbers:
45 is the greatest.

INPUT

45 26

QUESTIONS 8 / 8 completed

Character

CHARACTER

Write a program that takes a character as input and determines whether it is a vowel or a consonant.

PROGRAM EDITOR

C

Run

Save

INPUT

A

```
1 #include <stdio.h>
2 int main()
3 {
4     char ch;
5     printf("Enter a character: \n");
6     scanf("%c", &ch);
7     if (ch== 'a' || ch== 'e' || ch== 'i' || ch== 'o' || ch== 'u' || ch== 'A' ||
8         ch== 'E' || ch== 'I' || ch== 'O' || ch== 'U')
9     {
10        printf("%c is a Vowel\n", ch);
11    }
12    else
13    {
14        printf("%c is a consonant\n", ch);
15    }
16    return 0;
17 }
```

OUTPUT

```
Enter a character:
A is a Vowel
```

QUESTIONS 8 / 8 completed

Age

AGE

Write a program that takes an age as input and determines whether the person is a child, teenager, adult, or senior citizen.

PROGRAM EDITOR

C

Run

Save

INPUT

25

```
1 #include <stdio.h>
2 int main()
3 {
4     int age;
5     printf("Enter age: \n");
6     scanf("%d", &age);
7     if (age < 13)
8         printf("Child\n");
9     else if (age >= 13 && age <= 19)
10        printf("Teenager\n");
11     else if (age >= 20 && age <= 59)
12        printf("Adult\n");
13     else
14        printf("Senior Citizen\n");
15    return 0;
16 }
```

OUTPUT

```
Enter age:
Adult
```



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QUESTIONS 8 / 8 completed

Temperature

TEMPERATURE

Write a program that takes a temperature as input and determines whether it is hot, warm, or cold.

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 int main()
3 {
4     float temp;
5     printf("Enter Temperature (c): \n");
6     scanf("%f", &temp);
7     if (temp >= 30)
8         printf("Hot\n");
9     else if (temp >= 20)
10        printf("Warm\n");
11    else
12        printf("Cold\n");
13    return 0;
14 }
```

OUTPUT

Enter Temperature (c):
Warm

INPUT

34



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QUESTIONS 8 / 8 completed

Ups-Lows

UPS-LOWS

Write a program that takes a character as input and determines whether it is uppercase or lowercase.

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 int main()
3 {
4     char ch;
5     printf("Enter a character: \n");
6     scanf("%c", &ch);
7     if (ch >= 'A' && ch <= 'Z')
8         printf("%c is an uppercase letter.\n", ch);
9     else if (ch >= 'a' && ch <= 'z')
10        printf("%c is a lowercase letter.\n", ch);
11    else
12        printf("%c is not an alphabet.\n", ch);
13    return 0;
14 }
```

OUTPUT

Enter a character: G is an uppercase letter.

INPUT

G

QUESTIONS 8 / 8 completed

Multiple

MULTIPLE

Write a program that takes a number as input and determines whether it is a multiple of 3 or 5.

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 int main()
3 {
4     int num;
5     printf("Enter a number: \n");
6     scanf("%d", &num);
7     if (num % 3 == 0 && num % 5 == 0)
8         printf("%d is a multiple of both 3 and 5.\n", num);
9     else if (num % 3 == 0)
10        printf("%d is a multiple of 3.\n", num);
11     else if (num % 5 == 0)
12        printf("%d is a multiple of 5.\n", num);
13     else
14        printf("%d is not a multiple of 3 or 5.\n", num);
15     return 0;
16 }
```

INPUT

15

OUTPUT

Enter a number: 15 is a multiple of both 3 and 5.

QUESTIONS 8 / 8 completed

Grade

GRADE

Write a program that takes a grade as input and determines the corresponding letter grade (A, B, C, D, or F).

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 int main()
3 {
4     int grade;
5     printf("Enter the marks (0-100): \n");
6     scanf("%d", &grade);
7     if (grade >= 90 && grade <= 100)
8         printf("Grade = A\n");
9     else if (grade >= 80)
10        printf("Grade = B\n");
11     else if (grade >= 70)
12        printf("Grade = C\n");
13     else if (grade >= 60)
14        printf("Grade = D\n");
15     else if (grade >= 0)
16        printf("Grade = F\n");
17     else
18        printf("Invalid Marks\n");
19     return 0;
20 }
```

INPUT

75

OUTPUT

Enter the marks (0-100):
Grade = C

Looping



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QUESTIONS 18 / 18 completed

Sum(ODD,EVEN) ▾

SUM(ODD,EVEN)

C Program to Find the Sum of Even and Odd Numbers.
Problem Description
The program takes the number N and finds the sum of odd and even numbers from 1 to N.

Testcases:
-5
10
4
100

PROGRAM EDITOR

C ▾

Run Save

```
1 #include <stdio.h>
2 int main()
3 {
4     int N, i;
5     int oddSum = 0, evenSum = 0;
6     printf("Enter the value of N: \n");
7     scanf("%d", &N);
8     for (i = 1; i <= N; i++)
9     {
10         if (i % 2 == 0)
11             evenSum = evenSum + i;
12         else
13             oddSum = oddSum + i;
14     }
15     printf("Sum of Even Numbers = %d\n", evenSum);
16     printf("Sum of Odd Numbers = %d\n", oddSum);
17     return 0;
18 }
```

OUTPUT

Enter the value of N:
Sum of Even Numbers = 650
Sum of Odd Numbers = 625

INPUT

50



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QUESTIONS 18 / 18 completed

Natural ▾

NATURAL

Write a program that prints the first n natural numbers using a for loop.

PROGRAM EDITOR

C ▾

Run Save

```
1 #include <stdio.h>
2 int main()
3 {
4     int n, i;
5     scanf("%d", &n);
6     for (i = 1; i <= n; i++)
7     {
8         printf("%d ", i);
9     }
10     return 0;
11 }
```

OUTPUT

1 2 3 4 5

INPUT

5



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QUESTIONS 18 / 18 completed

Sum

SUM

Write a program that calculates the sum of the first n natural numbers using a while loop.

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 int main()
3 {
4     int n, i = 1, sum = 0;
5     scanf("%d", &n);
6     while (i <= n)
7     {
8         sum = sum + i;
9         i++;
10    }
11    printf("Sum of first %d natural numbers = %d\n", n, sum);
12    return 0;
13 }
```

OUTPUT

Sum of first 10 natural numbers = 55

INPUT

10



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QUESTIONS 18 / 18 completed

Table

TABLE

Write a program that prints the multiplication table of a given number using a for loop.

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 int main()
3 {
4     int n, i;
5     printf("Multiplication Table of %d:\n", n);
6     scanf("%d", &n);
7     for (i = 1; i <= 10; i++)
8         printf("%d x %d = %d\n", n, i, n * i);
9     return 0;
10 }
```

INPUT

2

QUESTIONS 18 / 18 completed

Prime

PRIME

Write a program that takes a number as input and determines whether it is a prime number or not using a while loop.

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 int main()
3 {
4     int n, i = 2, prime = 1;
5     printf("Enter a number: \n");
6     scanf("%d", &n);
7     if (n <= 1)
8     {
9         prime = 0;
10    }
11    else
12    {
13        while (i <= n / 2)
14        {
15            if (n % i == 0)
16            {
17                prime = 0;
18            }
19            i++;
20        }
21    }
```

OUTPUT

Enter a number:
13 is a prime Number.

INPUT

13

QUESTIONS 18 / 18 completed

Count Vows

COUNT VOWS

Write a program that takes a string as input and counts the number of vowels in the string using a for loop.

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 int main()
3 {
4     char str [100];
5     char ch;
6     int i, count = 0;
7     printf("Enter a String: \n");
8     scanf("%[^\n]", str);
9     for (i = 0; str[i] != 0; i++)
10    {
11        ch = str[i];
12        if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' ||
13            ch == 'u' || ch == 'A' || ch == 'E' || ch == 'I' ||
14            ch == 'O' || ch == 'U')
15            count++;
16    }
17    printf("Number of vowels = %d", count);
18    return 0;
19 }
```

OUTPUT

Enter a String:
Number of vowels = 2

INPUT

Hello



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QUESTIONS 18 / 18 completed

Fibonacci

FIBONACCI

Write a program that prints the Fibonacci series up to a given number using a while loop.

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 int main()
3 {
4     int n, i, first = 0, second = 1, third = 0;
5     scanf("%d", &n);
6     for (int i = 1; i <= n; i++){
7         third = first + second;
8         printf("%d ", first);
9         first = second;
10        second = third;
11    }
12    return 0;
13 }
```

OUTPUT

```
0 1 1 2 3 5 8 13 21 34
```

INPUT

10



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QUESTIONS 18 / 18 completed

Factorial

FACTORIAL

Write a program that takes a number as input and prints its factorial using a for loop.

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 int main()
3 {
4     int n;
5     long long factorial = 1;
6     scanf("%d", &n);
7     for (int i = 1; i <= n; i++) {
8         factorial *= i;
9     }
10    printf("Factorial of %d = %lld\n", n, factorial);
11    return 0;
12 }
```

OUTPUT

```
Factorial of 5 = 120
```

INPUT

5

QUESTIONS 18 / 18 completed

Pyramid

PYRAMID

Write a program that prints a pyramid pattern of stars using nested loops.

PROGRAM EDITOR

C

Run

Save

INPUT

5

```
1 #include <stdio.h>;
2 int main()
3 {
4     int i, j, n;
5     scanf("%d", &n);
6     for (i = 1; i <= n; i++){
7         for (j = 1; j <= n - i; j++){
8             printf(" ");
9         }
10        for (j = 1; j <= 2 * i - 1; j++){
11            printf("*");
12        }
13        printf("\n");
14    }
15    return 0;
16 }
```

OUTPUT

```
  *
 ***
*****
*****
*****
```

QUESTIONS 18 / 18 completed

Diamond

DIAMOND

Write a program that prints a diamond pattern of stars using nested loops.

PROGRAM EDITOR

C

Run

Save

INPUT

5

```
1 #include <stdio.h>
2 int main()
3 {
4     int i, j, n;
5     scanf("%d", &n);
6     for (i = 1; i <= n; i++){
7         for (j = 1; j <= n - i; j++){
8             printf(" ");
9         }
10        for (j = 1; j <= 2 * i - 1; j++){
11            printf("*");
12        }
13        printf("\n");
14    }
15    for (i = n - 1; i >= 1; i--){
16        for (j = 1; j <= n - i; j++){
17            printf(" ");
18        }
19        for (j = 1; j <= 2 * i - 1; j++){
20            printf("*");
21        }
22    }
```

OUTPUT

```
  *
 ***
*****
*****
*****
***
  *
```

QUESTIONS 18 / 18 completed

R- triangle

R- TRIANGLE

Write a program that prints a right-angled triangle pattern of stars using nested loops.

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 int main()
3 {
4     int i, j, n;
5     scanf("%d", &n);
6     for (i = 1; i <= n; i++){
7         for (j = 1; j <= i; j++){
8             printf("*");
9         }
10        printf("\n");
11    }
12    return 0;
13 }
```

INPUT

5

OUTPUT

```
*
**
***
****
*****
```

QUESTIONS 18 / 18 completed

Hollow Square

HOLLOW SQUARE

Write a program that prints a hollow square pattern of stars using nested loops.

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 int main()
3 {
4     int i, j, n;
5     scanf("%d", &n);
6     for (i = 1; i <= n; i++) {
7         for (j = 1; j <= n; j++) {
8             if (i == 1 || i == n || j == 1 || j == n)
9                 printf("*");
10            else
11                printf(" ");
12        }
13        printf("\n");
14    }
15    return 0;
16 }
```

INPUT

4

OUTPUT

```
****
*  *
*  *
****
```

QUESTIONS 18 / 18 completed

PROGRAM EDITOR

C

Run

Save

INPUT

Hollow Diamond

HOLLOW
DIAMOND

Write a program that prints a hollow diamond pattern of stars using nested loops.

```
1 #include <stdio.h>
2 int main()
3 {
4     int i, j, n;
5     scanf("%d", &n);
6     for (i = 1; i <= n; i++) {
7         for (j = 1; j <= n - i; j++) {
8             printf(" ");
9         }
10        for (j = 1; j <= 2 * i - 1; j++) {
11            if (j == 1 || j == 2 * i - 1 || i == n)
12                printf("*");
13            else
14                printf(" ");
15        }
16        printf("\n");
17    }
18    for (i = n - 1; i >= 1; i--) {
19        for (j = 1; j <= n - i; j++) {
20            printf(" ");
21        }
22    }
```

OUTPUT

```
  *
 * *
* * *
*****
* * *
 * *
  *
```

5

QUESTIONS 18 / 18 completed

PROGRAM EDITOR

C

Run

Save

INPUT

Pascal's triangle

PASCAL'S
TRIANGLE

Write a program that prints a Pascal's triangle pattern using nested loops.

```
1 #include <stdio.h>
2 int main() {
3     int rows, coef = 1, space, i, j;
4     scanf("%d", &rows);
5     for (i = 0; i < rows; i++) {
6         for (space = 1; space <= rows - i; space++) {
7             printf(" ");
8         }
9         for (j = 0; j <= i; j++) {
10            if (j == 0 || i == 0)
11                coef = 1;
12            else
13                coef = coef * (i - j + 1) / j;
14            printf("%4d", coef);
15        }
16        printf("\n");
17    }
18    return 0;
19 }
```

OUTPUT

```
      1
     1 1
    1 2 1
   1 3 3 1
  1 4 6 4 1
```

5

QUESTIONS 18 / 18 completed

zigzag

ZIGZAG

Write a program that prints a zigzag pattern of stars using nested loops.

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 int main()
3 {
4     int i, j, n;
5     printf("Enter the number of cols: \n");
6     scanf("%d", &n);
7     for(i = 0; i < 3; i++)
8     {
9         for(j = 0; j < n; j++)
10        {
11            if((i == 0 && j % 4 == 2) ||
12               (i == 1 && (j % 4 == 1 || j % 4 == 3)) ||
13               (i == 2 && j % 4 == 0))
14            {
15                printf("**");
16            }
17            else
18            {
19                printf(" ");
20            }
21        }
22    }
```

OUTPUT

```
Enter the number of cols:
* * *
* * * * *
* * * *
```

INPUT

13

QUESTIONS 18 / 18 completed

Spiral pattern

SPIRAL PATTERN

Write a program that prints a spiral pattern of numbers using nested loops.

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 int main()
3 {
4     int n;
5     scanf("%d", &n);
6     int top = 0, bottom = n - 1, left = 0, right = n - 1;
7     int i, j, num = 1;
8     int a[n][n];
9     while (top <= bottom && left <= right){
10         for (j = left; j <= right; j++)
11             a[top][j] = num++;
12         top++;
13         for (i = top; i <= bottom; i++)
14             a[i][right] = num++;
15         right--;
16         if (top <= bottom){
17             for (j = right; j >= left; j--){
18                 a[bottom][j] = num++;
19             }
20             bottom--;
21         }
```

OUTPUT

```
1  2  3  4  5
16 17 18 19 6
15 24 25 20 7
14 23 22 21 8
13 12 11 10 9
```

INPUT

5



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QUESTIONS 18 / 18 completed

Armstrong

ARMSTRONG

Write a program to find the given number is Armstrong or Not.

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 int main()
3 {
4     int n, temp, rem, sum = 0;
5     scanf("%d", &n);
6     temp = n;
7     while (temp != 0){
8         rem = temp % 10;
9         sum = sum + rem * rem * rem;
10        temp = temp / 10;
11    }
12    if (sum == n)
13        printf("Armstrong Number");
14    else
15        printf("Not an Armstrong Number");
16    return 0;
17 }
```

OUTPUT

Armstrong Number

INPUT

153



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QUESTIONS 18 / 18 completed

Perfect

PERFECT

Write a program to find the given number is Perfect number or Not.

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 int main()
3 {
4     int n, i, sum = 0;
5     scanf("%d", &n);
6     for (i = 1; i < n; i++){
7         if (n % i == 0){
8             sum = sum + i;
9         }
10    }
11    if (sum == n)
12        printf("Perfect Number");
13    else
14        printf("Not a Perfect Number");
15    return 0;
16 }
```

OUTPUT

Perfect Number

INPUT

28